

Rocco Nardo

COSC 481 Capstone

COFSE Website Redesign and Development Final Report

Overall Progress:

Since the midterm, I have completed several key objectives to ensure that the site became functional. I developed backend functionality to send COFSE membership applications to the appropriate administrator, where all user inputs can be reviewed and approved or declined by clicking a button. I also completed a process that automatically converts approved applicants into site members as the user will receive an email prompting them to set their password. In addition, I worked with the Wix Members app to create a designated page where members can log in and update their information if needed. I also created and designed members-only subpages that contain exclusive resources only accessible to site members. While not originally an objective, I also developed an automated process to import current COFSE members into the application database and automatically generate site contacts, simplifying the process of bringing over current members. Additionally, I worked with Dr. Ferrara to establish a new domain hosted through Wix and connect it to the new site. Lastly, I created detailed documentation and commented all code to ensure that COFSE administrators can easily maintain the site once the semester is over. Overall, the project is largely complete, with most desired features implemented and functioning.

Remaining Work:

While the project is largely complete and the site is live and functional, there are still two remaining objectives. First, the integration of a mass email communication platform like Constant Contact still needs to be done as I did not have enough time to complete it. In addition,

although I created contacts for all existing COFSE members from the old site, invitations still need to be sent through the Wix Dashboard to grant them membership on the new site. Once the invitations are sent and the email integration is implemented, the site will be fully complete and operational.

Unanticipated Challenges:

Live Site Permissions – Once I published the site, I found that several tasks that could be done in “Preview” mode now failed. By examining the error messages in the console, I found that the issues were caused by stricter default permission settings applied in the published version for security purposes. Through researching the specific errors I was receiving, I discovered that users were not granted sufficient access to submit applications, which automatically store their input into the database. As a result, I adjusted my code to ensure that users are granted the minimum level of access required to submit application data.

Importing Current Members – While I began working on importing current COFSE members to the new site, I found that there was no easy way provided by Wix to do so. Therefore, I worked with Dr. Ferrara to determine how COFSE was currently storing its member information and learned that it was maintained in an Excel spreadsheet. Using this file, I imported the existing member data into the same Wix database that holds membership applications to ensure all user information is stored in the same place. After completing the import, I decided to develop an automated process to convert the database records into Wix contacts, simplifying the process of bringing over current members.

Transferring the COFSE Domain – When preparing to migrate the domain from the existing site to new site, COFSE administrators discovered that they were locked out of their domain account. Once they discovered that it would take sixty days to get access to the domain, we decided to

establish a new domain name through Wix and connect it to the site to ensure the project remained on schedule.

Objectives / Design Changes:

While continuing the work on the site, I made minor changes to the style of the site at the request of the COFSE executive board. I increased the size of the font, included more navigation buttons, and added more pictures. I also hid the Links and Member Directory pages, as they are currently blank and not being used.

There were also several changes to the objectives. First, we decided to remove the objective of implementing the Wix Payment feature due to its cost. As a result, we decided to handle membership dues through links to the COFSE Venmo and PayPal accounts. In addition, while we did not fully remove the objective of implementing Constant Contact, we made it a lesser priority than the other objectives. Since implementing Constant Contact relies on having site members sync to it, Dr. Ferrara and I realized that I needed to put more time into finishing the member creation process before we could start working on Constant Contact. While I did not have time to begin its implementation, it remains a feature that could possibly be added in the future. Lastly, we added an objective of importing members of the existing site to the new site. Once the membership approval process was complete, we decided that importing existing members was more efficient than requiring everyone to reapply on the new site.

Contributions:

As I am the only student working on this project, I have worked on all the completed tasks with 100% of the effort. These include developing the COFSE member approval process, handling member login setup and updates, creating and designing members-only pages that have restricted access, importing current members into the new site, establishing and connecting a new domain,

and creating detailed documentation. I have worked on this project for a total of **58.25 hours** since the midterm, for an average of **8.3** hours per week over the last seven weeks.

Future Work:

As the semester is coming to an end, Dr. Ferrara and the COFSE administrators will take over maintaining and working on the site. First, they should analyze the documentation to fully understand where changes to the code should be made, as well as learn the different Wix tools that have been used. Once they understand how the site works, they should send invites to all the contacts of the site who have been members of the existing site. They should also decide if they still want to integrate the site with Constant Contact or not. If they do, they can use a platform called Zapier to automatically add Wix members to their Constant Contact account. Lastly, COFSE administrators should continually use Wix Editor and Wix Velo to maintain and update the site by adding current content, keeping the design modern, and adding needed fields to the application forms.

Code Repository:

<https://github.com/rocco-nardo/COFSE-Capstone.git>

Dr. Ferrara has approved of making the repository public.

Work Log:

Rocco Nardo Computer Science Capstone Work Log			
Task	Details	Time For Task (h)	TOTAL TIME (h)
Meeting with Dr. Ferrara (1/20/26)	Discussed the COFSE project more in-depth, negotiated points of the design phase, and planned future meetings	0.50	107.5
Explore Current COFSE Website (1/20/26)	Sign up as COFSE member and explore current website familiarizing myself with its content and current shortcomings	0.75	

Research the features of Wix platform and Wix Velo - custom coding feature (1/22/26)	Research what features Wix provides, as well as how I can add custom styling and behavior through custom coding instead of relying on their Wix Editor	1.50	
Examine current site to lay out page/navigation of new site (1/26/26)	Examine the content on each page of current site and consolidate/add pages to improve simplicity and understanding	0.75	
Research specific features that Dr. Ferrara requested (1/26/26)	Research the best ways to implement specific features in Wix, like accepting payment, constant contact, and members only pages to ensure they can be done in a reasonable amount of time	1.00	
Meeting with Dr. Ferrara (1/29/26)	Review my Design and Objectives Report, make various updates, and set up Wix account and environment	0.50	
Examine and look through Wix provided templates (1/31/26)	Look through Wix templates for various types of sites to gain inspiration for my new site design	0.75	
Set up Wix environment for my site (2/2/26)	Create a site on the Wix platform, provide Dr. Ferrara with access to site, and fill out various settings to ensure the site is in demo mode and there will be no cost yet	0.50	
Create navigation bar and a blank page for every agreed upon page (2/2/26)	Use Wix platform to create navigation bar, create blank pages, and create subpages/dependencies (Wix requires this to be done in the editor instead of coding console)	0.75	
Find and download all content from current site that will be needed (2/3/26)	Examine current site to find and download content like images, PDFs, and website links that will be needed in the new site	1.25	
Research, learn, and familiarize myself with the features of Wix Velo (2/3/26)	Research to learn what coding features Wix Velo provides and does not provide (Can not create HTML from scratch but can use JavaScript for behavior and some CSS features). Learn the different tools of Wix Velo and practice basic JavaScript	3.00	
Add content to and design Home Page (2/4/26)	Used both Wix Editor and Wix Velo to add the up-to-date content to the new site and then design the page in a clean and modern way. (Took longer than expected as it took a while to learn how to handle hero images and create links using Velo)	2.00	
Add content to and design About Us Page (2/4/26)	Use both Wix Editor and Wix Velo to add the up-to-date content to the new page and design in	0.75	

Meeting with Dr. Ferrara (2/5/26)	Show Dr. Ferrara the beginnings of the new site set up and discuss and clarify what content is expected to be transported to the new site	0.5	
Add images of executive board and bio pages as Dr. Ferrara requested (2/5/26)	Add a picture of each executive board member along with their contact info, and then a button to each individual member's page which includes their bio. Took longer than expected as I had to learn how to use the Wix-location library to link from one page of the site to another using Velo	2	
Add content to and design the Committees page and subpages (2/10/26)	Use both Wix Editor and Wix Velo to add the up-to-date content to the new page and design it	1.25	
Add content to and design the Journal page (2/12/26)	Use both Wix Editor and Wix Velo to add the up-to-date content to the new page and design it; Took time as there were a lot of links in this page	1.75	
Add content to and design the DDE main page (2/12/26)	Use both Wix Editor and Wix Velo to add the up-to-date content to the new page and design it	0.75	
Migrate all the current DDE chapters to the new site (2/13/26)	Migrate all DDE chapters to the new chapters page on the site. Had to make each chapter and its elements its own <p> element to make the email linking easier to it took a long time as there are many chapters	2.5	
Loop through all text elements of DDE chapters page to create links for every email (2/16/26)	I began using .html feature in Velo to create links for each email but it was taking too long. Spent some time figuring out how to loop through the <p> elements searching for email patterns, adding links automatically to all emails	2	
Meeting with Dr. Ferrara (2/17/26)	Show Dr. Ferrara the new website with most of the content and pages migrated. Ask questions regarding specific content	0.5	
Connect Wix site with GitHub (2/18/26)	Connect my Wix site with GitHub and VS code and test that connection works properly	1	
Update Styling Code (2/19/26)	Update the code of several pages that handle styling	1.5	
Continue to migrate content and create link for a variety of different pdf files (2/23/26)	I ran into several issues with creating the links as many of the links did not show the expected text. I also spent time attempting to change the URL of the pdf to make it easier to understand. It also took a long time as	3	

	there were many different pdf files that needed a link		
Attempt to write code to create the form applications (2/24/26)	My first attempt to create forms was not making much progress as I was running into several errors, so I researched the best way to do it in WIX	2	
Create the design of the membership application on Wix (2/25/26)	Create the layout of the membership application with the correct input types on WIX	1.5	
Meeting with Dr. Ferrara (2/26/26)	Show Dr. Ferrara progress with designing the pages and providing links to everything. We then discussed how I should go about the input submission forms	0.5	
Begin working on code to send input to correct email (2/26/26)	Began working in VS code to send the info input by user to correct email address	2	
Research Wix Payments platform feature (3/2/26)	Research Wix Payments platform feature to see how it works	0.75	
Set up base membership page (3/5/26)	Create and set up base membership page that also links to pages to login or apply as a member	1.25	
Fix some styling (3/5/26)	Fix styling in Velo on the committee page	0.5	
Setup Design for DDE Application (3/6/26)	Design and set up the inputs in Wix for the DDE membership application	1.25	
Setup up database for storing DDE submissions and create a triggered email 3/8/26	Create a collection / database in Wix with fields to store each DDE submission and create a triggered email that will be sent to admin for each submission containing all the inputs	1.75	
Write frontend/backend code to send submissions to database and email to admin (3/8/26)	Add code to frontend page file to send submission to proper database when submit is clicked. Create a backend file that will then send the triggered email to admin containing all the input for each submission	2	
Fix design of triggered email for DDE (3/9/26)	Improve the design of the triggered email containing input of DDE submission	0.5	
Setup database for storing COFSE membership submission and create a triggered email (3/9/26)	Create a collection / database in Wix with fields to store each membership submission and create a triggered email that will be sent to admin for each submission containing all the inputs	1.5	

Create the base of admin review page (3/9/26)	Start with the creation of an admin review Wix page when admin can review and approve membership applications	0.25	
Meeting with Dr. Ferrara (3/10/26)	Showed Dr. Ferrara my progress and completion of saving and sending inputs of the DDE submission and tested that it worked. We then discussed how I will modify this progress for the COFSE submission	0.5	
Modify DDE Submission (3/10/26)	Based on input from the president of DDE, I changed some of the inputs of the form and database	0.5	
Add comments to code (3/11/26)	Add comments to code files to make it easier to understand	0.5	
Review current code and make clarifying modifications (3/12/26)	Look through all my current code and add comments/ clarifying modifications to make it easier to understand	1.25	
Create frontend and backend files that submit COFSE membership (3/13/26)	Create backend file and modify frontend membership application page to save COFSE membership submissions to collection and send the inputs to specific email	1.5	
Modify Triggered Email buttons for COFSE membership review (3/15/26)	Modify the triggered email so clicking on the review buttons redirects admin to page to approve/reject	0.75	
Add code to Admin Review page and backend files to bring functionality to the review membership page (3/17/26)	Add code to Admin Review page and modify cofseEmail page so the admin clicking approve/decline changes the review status field in the database - time spent for debugging	1.75	
Publish site with free, Wix given domain for testing (3/18/26)	Publish site (hidden so others can't find it) so I could test approval functionality, finding several problems once site went live	1	
Debug errors from live, published site (3/18/26)	Debugged and fixed some errors that occurred only once site went live	1.5	
Improved Admin Review page functionality (3/18/26)	Modified Admin Review to improve design and fix some small errors	1.75	
Meeting with Dr. Ferrara (3/19/26)	Meet with Dr. Ferrara to show progress and ask clarifying questions about how the membership works	0.5	
Fix errors relating to adding contacts based on submission (3/19/26)	Modify backend files to ensure new contacts were being created for admin receiving	2	

	emails as well as all users that submit a COFSE membership application		
Create all the different COFSE members only page and work on Annual Meetings page (3/22/23)	Create all the pages and use .html to add all the needed links in the page	1.5	
Work on Presentation and Additional Resources page (3/23/26)	Use .html to add all the different links needed to the pages	2	
Work on Workshop page (3/24/26)	Use .html to add all the different links needed to the pages	1.5	
Continue working on membership approval (3/25/26)	Continue to work on the functionality of membership approval	1.25	
Meeting with Dr. Ferrara (3/26/26)	Meet with Dr. Ferrara to show progress and ask clarifying questions about how the members only pages	0.5	
Continue working on membership approval (3/28/26)	Continue to work on the functionality of membership approval automatically making approved members a site member	2.5	
Work on emailing members that they have been approved (3/28/26)	Trouble shoot and eventually figure out how to email members that they have been approved	2	
Research how to create login in/sign up Wix site pages (3/30/26)	Research the best way to create a log in page for members to use to log in to the Wix site	1.25	
Work on creating login page for members (3/30/26)	Using different Wix techniques, try and debug ways until login page successfully works	3.25	
Work on access control for members only page (3/31/26)	attempt to use Wix velo to create access control but it was not working correctly so I added access control through Wix directly	1.75	
Improve Members-only functionality (4/1/26)	Improve members only functionality by making it easier to use and removing unneeded pages	1	
Add committee member info (4/6/26)	Based on executive board feedback, I added members of each committee, as well as link to their emails	1.5	
Make minor changes to site content (4/6/26)	Based on executive board feedback, make minor changes to some of the site content	0.5	
Update pictures and trouble shoot blurriness (4/7/26)	Replace pictures with new pictures and try different techniques to limit the blurriness	1	

Research and try different techniques to link site to constant contact (4/8/26)	Examine different techniques based on practicality and cost to link Wix site members to constant contact account - must discuss with Dr. Ferrara	2	
Meeting with Dr. Ferrara (4/9/26)	Meet with Dr. Ferrara to discuss process and how we plan on bringing over current members. Also discussed more about some of the modifications the executive board wanted	0.5	
Create buttons to improve functionality (4/9/26)	Create a variety of buttons like back buttons and buttons to go to other pages to improve navigation	0.75	
Modify page styling (4/11/26)	Based on feedback from executive board, I modified some of the styling of all the pages	0.75	
Add fields to application (4/12/26)	Based on feedback from executive board, I added fields to application and modified backend to handle them	0.5	
Import current members into the database for COFSE members (4/12/26)	Research the best way to import members into CMS, modify format of the Excel file so it can be imported, and test that everything imported properly	1.5	
Create automated process to automatically create contacts for all new member imports to CMS (4/13/26)	Create and modify files to automatically make contacts for each user added to the CMS because they are already members. Ran into trouble and took longer than expected	4	
Run the automated process to create contacts for members (4/13/26)	Run the automated process in batches of 25 and ensure each batch worked properly	1	
Add another application field (4/14/26)	Attempted to add a document input, but it was not working properly, so added text input	1	
Meeting with Dr. Ferrara (4/16/26)	Meet with Dr. Ferrara to discuss any other small modifications needed as well as brainstorm how to get domain as the domain account has restricted COFSE access	0.5	
Researched proper way to transfer domain (4/16/26)	Spent a lot of time learning to proper way to transfer domain from one site to another to ensure it is done safely	2	
Modify mobile design (4/16/26)	Use Wix Editor to improve design for mobile site	0.5	
Research how to properly add PayPal link without running into session problems (4/20/26)	The PayPal link I was trying to use kept breaking due to sessions so I researched how to get around it	0.75	
Meeting with Dr. Ferrara (4/21/26)	Meet with Dr. Ferrara to talk more about the domain and remaining work	0.5	

Add correct PayPal Link (4/21/26)	Add correct link so it would not break	0.25	
Improve comments on every file for documentation (4/24/26)	Add more descriptive and detailed comments	1.75	
Develop most of the documentation page (4/27/26)	Work on creating the documentation document to explain how everything works and what needs to be done in the future in a lot of detail	3.5	
Meeting with Dr. Ferrara (4/28/26)	Meet with Dr. Ferrara to explain most of the documentation phase. We also worked on adding the new domain name to the site	0.5	
Test site functionality after new domain (4/28/26)	Now that site was live on correct domain, I did an in-depth user test to ensure everything worked properly	1.25	
Update emails and URLs (4/28/26)	Now that site is on correct domain and tested fully, I modified emails to the appropriate administrator and corrected absolute links	0.25	
Work on getting site ready for administrators to work on (5/2/26)	I disconnected site from GitHub so COFSE can work directly in Wix Editor and Velo. I also removed all testing records in the databases	0.75	
Finish the documentation doc (5/3/26)	I added the last couple sections to the document, added it to GitHub, and linked it in my ReadMe	1.5	